A SHEAF WITH A REPRESENTABLE SURJECTIVE ETALE COVER BY AN ALGEBRAIC SPACE IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but complete: it produces the covering map U -> F and verifies the three properties, with no omitted step.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Composition of representable transformations. Lemma lemma-composition-representable-transformations, used at proof lines 1918-1921. Makes U -> F representable.
- Composition of representable transformations with a property. Lemma lemma-composition-representable-transformations-property, used at proof lines 1918-1921. Keeps surjectivity and etaleness under composition.
- Recognition criterion for a sheaf with a representable surjective etale cover. Lemma lemma-etale-locally-representable-gives-space (1884-1891, counted as L44182), used at proof lines 1922-1923. Concludes that F is an algebraic space.

Source reference. Original lemma 02WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3797–3797. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3807–3807. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0BGQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14428–14428. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let G be an algebraic space over S, let F be a sheaf on $(Sch/S)_{fppf}$, and let $G \to F$ be a representable transformation of functors which is surjective and étale. Then F is an algebraic space.*

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Pick a scheme U and a surjective étale morphism $U \to G$. Since G is an algebraic space $U \to G$ is representable. Hence the composition $U \to G \to F$ is representable, surjective, and étale. See Lemmas 02WA and 02WK. Thus F is an algebraic space by Lemma 0BGQ. □